

Problem 1) Given the data:

	x_1	x_2	y
0	1	2	1
1	2	5	1
2	3	4	-1
3	4	0	-1
4	0	1	-1

→ suppose we use this data to train an adaboost model with learning rate $L=1$ and obtain the three stumps as follows:

stump 1 : $I(x_1 < 2.5)$

stump 2 : $I(x_2 > 1.5)$

stump 3 : $I(x_2 > 4.5)$

a) complete the weight of the data in the first round

→ All observations have equal weights in the first round - since we have 5 observations, each one will have a weight of $1/5$ or 0.20

	x_1	x_2	y	w_1
0	1	2	1	0.20
1	2	5	1	0.20
2	3	4	-1	0.20
3	4	0	-1	0.20
4	0	1	-1	0.20

- b) After the first round, which observations should have the weights increased? Which should have the weights decreased? Why?

→ We generate the predictions of stump 1 after the first round. Stump 1 will predict 1 if $x_1 < 2.5$ and will predict -1 otherwise. The predictions of stump 1 are:

	x_1	x_2	y	y_1 (stump 1 prediction)
0	1	2	1	-1
1	2	5	1	1
2	3	4	-1	-1
3	4	0	-1	-1
4	0	1	-1	1

Since adaboost decreases weights for correctly classified observations and increases weights for misclassified observations, the weights will be increased for observation 4 and decreased for observations 0-3.

- c) Compute the weights of the data in the second and third round

→ calculate the errors ϵ_1 and voting power α_1 for stump 1. Since stump 1 has 1 misclassification at observation 4, the error of stump 1 is the weight of observation 4.

$$\text{So, } \epsilon_1 = 0.20 \quad \text{and} \quad \alpha_1 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right) = \frac{1}{2} \ln\left(\frac{.80}{.20}\right) = 0.69315$$

For the misclassified observation 4, the weight is updated as:

$$W_{\text{new}} = W_{\text{old}} \cdot e^{\alpha} = 0.20e^{0.69315} = 0.40$$

$$\text{For the remaining observations: } W_{\text{new}} = W_{\text{old}} \cdot e^{-\alpha} = 0.20e^{-0.69315} = 0.10$$

→ Need to normalize weights so they sum to 1. Divide weights by total ($0.40 \cdot 1 + 0.10 \cdot 4 = 0.80$), we get weights for the second round as follows →

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Problem 1 continued

c) $0.40/0.80 = 0.50$ and $0.10/0.80 = 0.125$

	x_1	x_2	y	y_1 (stump 1 pred.)	w_1	w_2
0	1	2	1	1	0.20	0.125
1	2	5	1	1	0.20	0.125
2	3	4	-1	-1	0.20	0.125
3	4	0	-1	-1	0.20	0.125
4	0	1	-1	1	0.20	0.50

moving onto stump 2 predictions: $\rightarrow$ *predicts 1 if $x_2 > 1.5$ and -1 otherwise *

	x_1	x_2	y	y_1 (stump 1 pred.)	w_1	w_2	y_2 (stump 2 pred.)
0	1	2	1	1	0.20	0.125	1
1	2	5	1	1	0.20	0.125	1
2	3	4	-1	-1	0.20	0.125	1
3	4	0	-1	-1	0.20	0.125	-1
4	0	1	-1	1	0.20	0.50	-1

$\rightarrow$ observation 2 is misclassified so that weight will increase and the rest will decrease

$$\epsilon_2 = 0.125 \quad \text{and} \quad \alpha_2 = L \cdot \frac{1}{2} \ln\left(\frac{1-0.125}{0.125}\right) = 0.972955$$

For misclassified observation 2, $w_{\text{new}} = w_{\text{old}} \cdot e^{\alpha} = 0.125 e^{0.972955} = 0.3307$

For remaining observations, $w_{\text{new}} = w_{\text{old}} \cdot e^{-\alpha} = 0.125 e^{-0.972955} = 0.04725$
and $0.50 e^{-0.972955} = 0.189$

Need to standardize so all sum to 1

$$\hookrightarrow (0.3307(1) + 0.04725(3) + 0.189(1)) = 0.66145$$

$$\frac{0.3307}{0.66145} = 0.50$$

$$\frac{0.04725}{0.66145} = 0.07143$$

$$\frac{0.189}{0.66145} = 0.28574$$

1C cont'd)

Now we have:

obs	x_1	x_2	y	y_1	w_1	w_2	y_2	w_3	y_3
0	1	2	1	1	0.20	0.125	1	.07143	-1
1	2	5	1	1	0.20	0.125	1	.07143	1
2	3	4	-1	-1	0.20	0.125	1	0.50	-1
3	4	0	-1	-1	0.20	0.125	-1	.07143	-1
4	0	1	-1	1	0.20	0.50	-1	0.28571	-1

* Stump 3 will predict 1 when $x_2 > 4.5$ and -1 otherwise *

Observation 0 is misclassified so $\epsilon_3 = 0.07143$

$$\text{and } \alpha_3 = L \cdot \frac{1}{2} \ln\left(\frac{1-\epsilon_3}{\epsilon_3}\right) = \frac{1}{2} \ln\left(\frac{1-0.07143}{0.07143}\right) = \underline{1.2825}$$

d) compute the voting power of the three stumps. Which stump has the highest voting power?

→ The errors and voting power of the three stumps are:

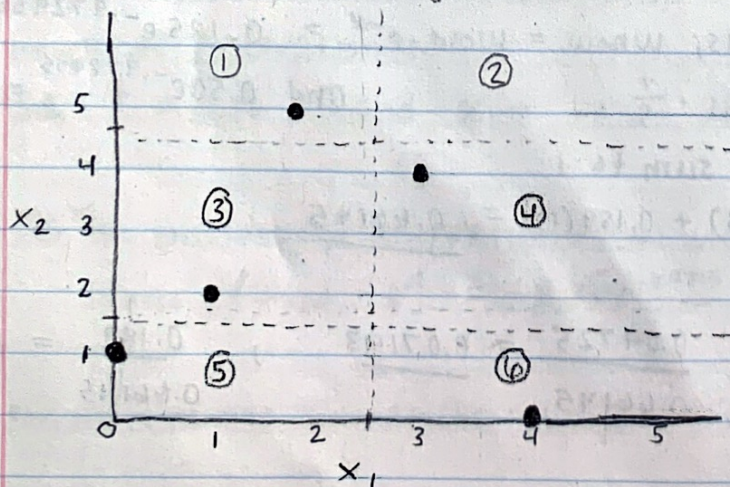
$$\epsilon_1 = 0.20 \quad \alpha_1 = 0.69315$$

$$\epsilon_2 = 0.125 \quad \alpha_2 = 0.972955$$

$$\epsilon_3 = 0.07143 \quad \alpha_3 = 1.2825$$

Stump 3 has the highest voting power

e) draw the decision boundary of the adaboost



* The 3 stumps break the coordinates into 6 regions *

$$\text{Sign} \left[.69315 I(x_1 < 2.5) + 0.972955 I(x_2 > 1.5) + 1.2825 I(x_2 > 4.5) \right] = A$$

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e cont'd) Region 1 $\begin{cases} x_1 < 2.5 \\ x_2 > 4.5 \end{cases} \rightarrow A = \text{sign}[0.69315(1) + 0.972955(1) + 1.2825(1)]$
 $= \text{sign}(2.9486) = 1 \text{ (Blue)}$

Region 2 $\begin{cases} x_1 > 2.5 \\ x_2 > 4.5 \end{cases} \rightarrow A = \text{sign}[0.69315(-1) + 0.972955(1) + 1.2825(1)]$
 $= \text{sign}[1.5623] = 1 \text{ (Blue)}$

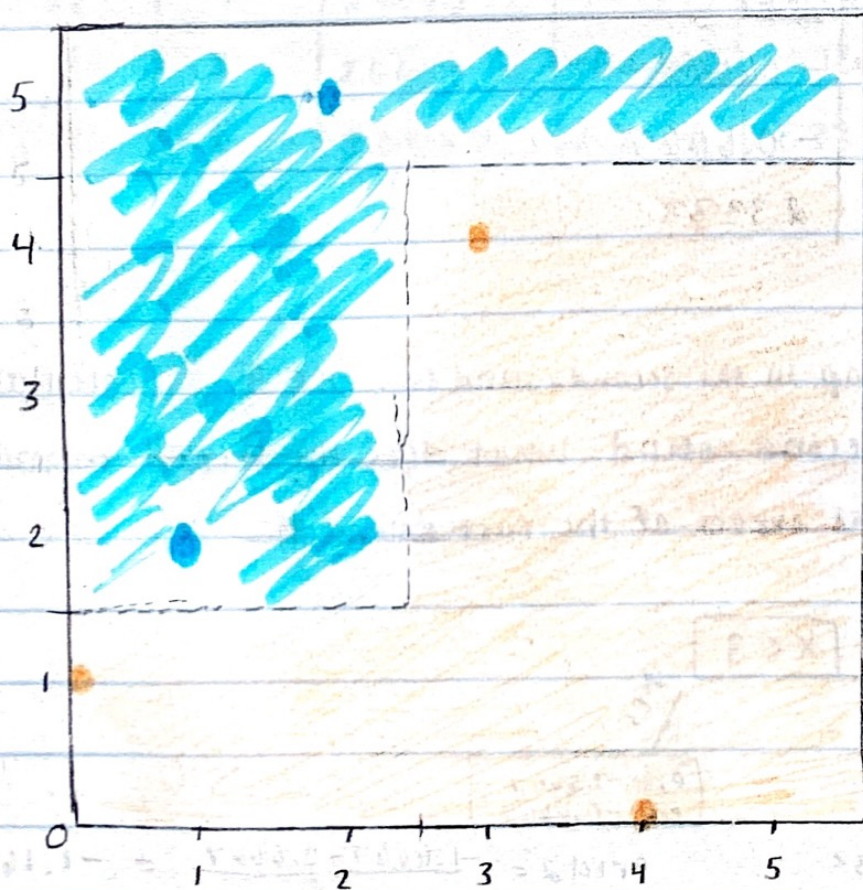
Region 3 $\begin{cases} x_1 < 2.5 \\ 1.5 < x_2 < 4.5 \end{cases} \rightarrow A = \text{sign}[0.69315(1) + 0.972955(1) + 1.2825(-1)]$
 $= \text{sign}(0.3836) = 1 \text{ (Blue)}$

Region 4 $\begin{cases} x_1 > 2.5 \\ 1.5 < x_2 < 4.5 \end{cases} \rightarrow A = \text{sign}[0.69315(-1) + 0.972955(1) + 1.2825(-1)]$
 $A = \text{sign}(-1.00) = -1 \text{ (red)}$

Region 5 $\begin{cases} x_1 < 2.5 \\ x_2 < 1.5 \end{cases} \rightarrow A = \text{sign}[0.69315(1) + 0.972955(-1) + 1.2825(-1)]$
 $= \text{sign}(-1.5623) = -1 \text{ (red)}$

Region 6 $\begin{cases} x_1 > 2.5 \\ x_2 < 1.5 \end{cases} \rightarrow A = \text{sign}[0.69315(-1) + 0.972955(-1) + 1.2825(-1)]$
 $= \text{sign}(\text{negative}) = -1 \text{ (red)}$

Decision Boundary (reds are negative and blues are positive)



1f) The misclassification error of the adaboost is 0 since the positive and negative points are perfectly classified within the decision boundary

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Exam 2.

Problem 2)

Given the data:

x	y
-1	1
2	2
4	5

We will train a gradient boosted tree (stump) on this data with the learning rate of 1.

a) calculate the prediction of the first round

→ prediction of round 1 is the average of y ($\bar{y}$)

$$\bar{y} = \frac{1+2+5}{3} = \frac{8}{3} = \boxed{2.6667}$$

x	y	pred 1
-1	1	2.6667
2	2	2.6667
4	5	2.6667

b) calculate the target of the second round

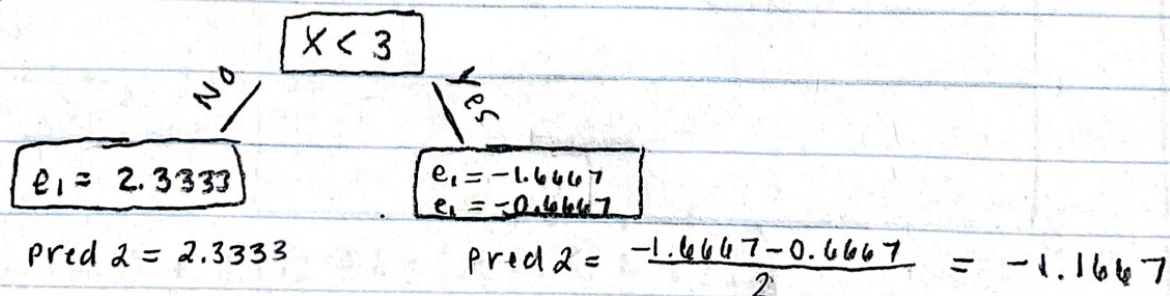
↳ target of the second round is the errors of the first round: $e_1 = y - \text{pred}_1$

x	y	pred 1	e_1
-1	1	2.6667	-1.6667
2	2	2.6667	-0.6667
4	5	2.6667	2.3333

c) suppose that the stump in the second round is: $x < 3$ calculate the predictions of the second round. What does the second round predict?

* second round predicts the errors of the first round *

Stump is $x < 3$



Exam 2

Problem 2 continued

c)

Table is as follows:

X	Y	Pred 1	e_1	Pred 2
-1	1	2.6667	-1.6667	-1.1667
2	2	2.6667	-0.6667	-1.1667
4	5	2.6667	2.3333	2.3333

d) What are the final predictions of the gradient boosting model for y in the training data after the second round?

→ sum of first & second round predictions

Therefore,

X	Y	Pred 1	Pred 2	Pred y
-1	1	2.6667	-1.1667	1.5
2	2	2.6667	-1.1667	1.5
4	5	2.6667	2.3333	5

e) calculate the target of the third round

↳ target of third round is error of second round (e_2) which is $e_1 - \text{Pred}_2$

X	Y	Pred 1	e_1	Pred 2	e_2
-1	1	2.6667	-1.6667	-1.1667	-0.5
2	2	2.6667	-0.6667	-1.1667	0.5
4	5	2.6667	2.3333	2.3333	0

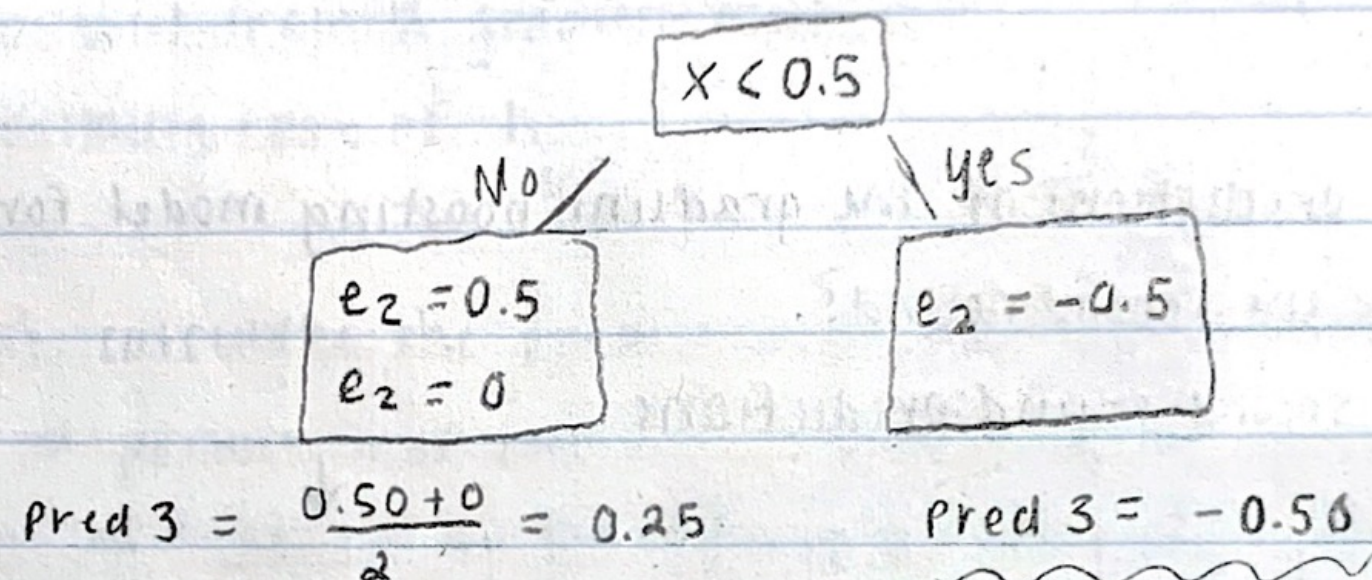
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Exam 2

- f) Suppose that the stump in the third round is $X < 0.5$. Calculate the predictions of the third round. What does the third round predict?

↪ Third round predicts the errors of the second round



This results as:

x	y	Pred 1	e_1	Pred 2	e_2	Pred 3
-1	1	2.6667	-1.6667	-1.1667	-0.50	-0.50
2	2	2.6667	-0.6667	-1.1667	0.50	0.25
4	5	2.6667	2.3333	2.3333	0	0.25

- g) What are the Final predictions of the gradient boosting model for y in the training data after the third round?

* predictions for y after the third round are the sum of predictions for rounds 1, 2, and 3

Results:

x	y	Pred 1	Pred 2	Pred 3	FINAL PRED
-1	1	2.6667	-1.1667	-0.50	1
2	2	2.6667	-1.1667	0.25	1.75
4	5	2.6667	2.3333	0.25	5.25

The final predictions are 1, 1.75 and 5.25 for observations 1, 2, and 3, respectively.